Experiment 05

Aim: To automate application deployment using GitHub/ Jenkins.

LO Mapped: LO3

LO: To design and configure Continuous Integration / Continuous Deployment pipelines using Jenkins, GitHub Actions and containerization tools.

Theory

This experiment demonstrates the integration of Jenkins with a GitHub repository to establish an automated Continuous Integration and Continuous Deployment (CI/CD) pipeline. The workflow tracks source code changes, retrieves files via Source Code Management (SCM), executes build and test stages, and deploys the web application locally. **Step-by-Step Implementation Details**

Step 1: Project Setup and Source Code Creation

- **Directory Structure Setup:** Created a dedicated project folder named JenkinsProject within the local workspace environment to isolate version control from broader directory bloat.
- **Web Asset Creation:** Developed a sample front-end file (index.html) containing an HTML5 structure and heading (<h1>Welcome to Tanmay's CI/CD Deployed Web Application!</h1>).
- **Pipeline Script Creation:** Drafted a declarative Jenkinsfile specifying three core stages: **Build**, **Test**, and **Deploy**, utilizing the agent any directive and Windows batch (bat) commands for deployment execution.

Step 2: Version Control and GitHub Integration

- **Repository Initialization:** Initialized a local Git repository inside the project folder using git init.
- **Staging and Committing:** Staged all project assets (git add .) and committed them with the message "Initial commit for Jenkins lab".
- **Branch Configuration & Push:** Renamed the default branch to main, linked the remote repository origin to [https://github.com/TanmayAhuja2006/jenkins-sampleapp.git](https://github.com/TanmayAhuja2006/jenkins-sample-app.git), and successfully pushed the codebase (git push -u origin main).

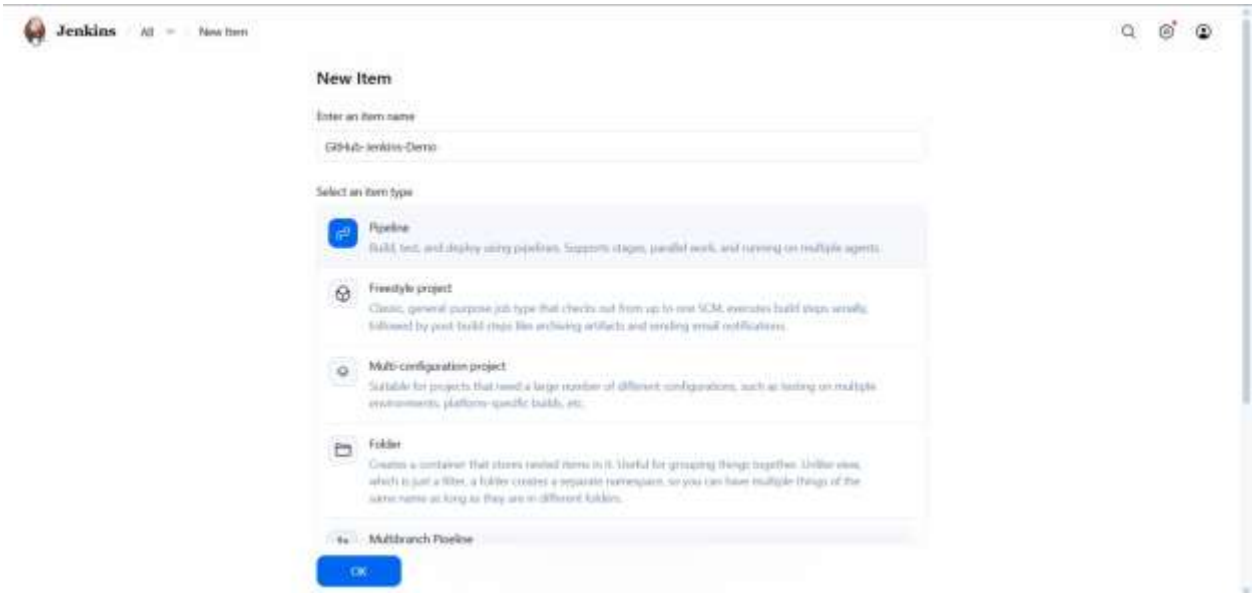
Step 3: Jenkins Job Configuration and Troubleshooting

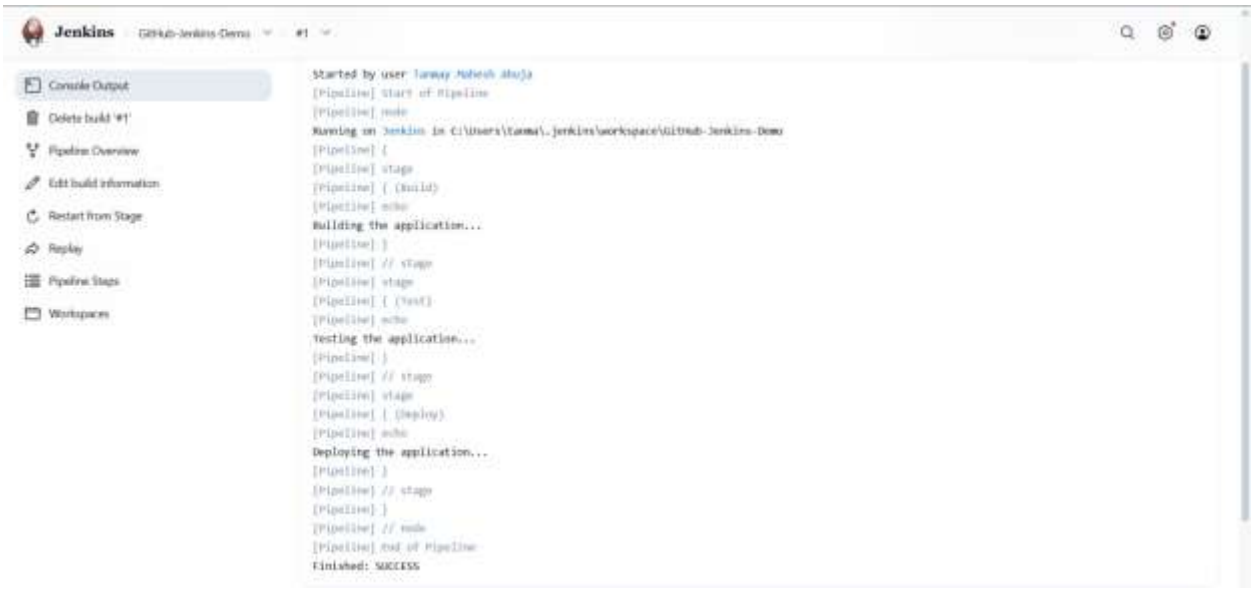
- **Pipeline Job Execution (#1 & #2):** Configured a Jenkins pipeline job named GitHubJenkins-Demo pointing to the GitHub repository. An initial trial resulted in Build #2 failing with the error ERROR: Unable to find Jenkinsfile from git.
- **Resolution:** Addressed the error caused by file naming conventions by ensuring the pipeline script was properly recognized as Jenkinsfile in the repository root.

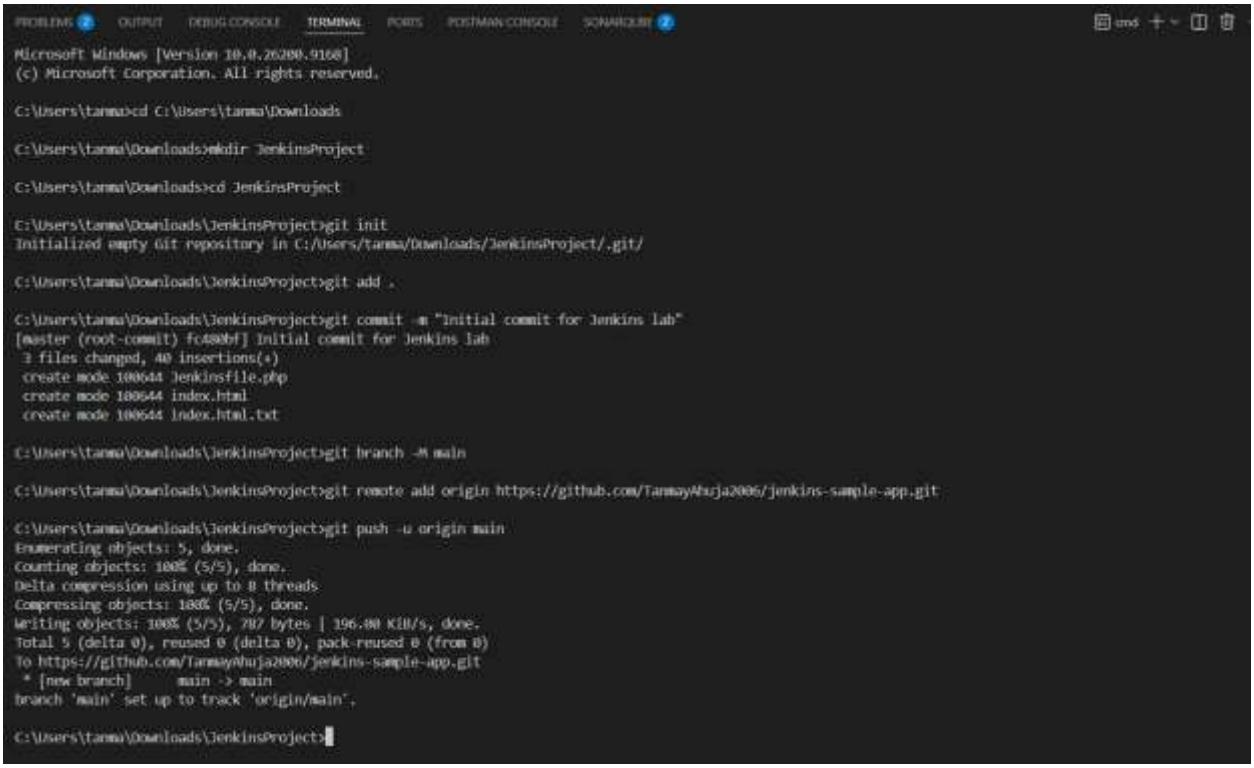
Step 4: Successful Pipeline Execution (Build #3)

- **SCM Checkout:** Triggered Build #3, which successfully cloned and fetched the remote repository (https://github.com/TanmayAhuja2006/jenkins-sample-app) referencing commit beec0df7703834acbb6571896709e7d3e017d294 on the main branch.
- **Build Stage:** Executed the first stage, logging the message: Building and preparing web assets....
- **Test Stage:** Executed validation routines, logging: Validating HTML structure....
- **Deploy Stage:** Executed local deployment steps via Windows batch commands:
    - Checked for the existence of target directory C:\Deployments\app and created it using mkdir if missing. o Copied index.html into the deployment directory (C:\Deployments\app\index.html), successfully copying 1 file.
- **Completion:** The pipeline finished with a status of SUCCESS.

Implementation :




Jenkins
GitHub-Jenkins-Demo
#1
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Started by user Tanmay Mahesh Ahuja
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\tanma\.jenkins\workspace\GitHub-Jenkins-Demo
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Testing the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying the application...
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS


PROBLEMS 2
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
POSTMAN CONSOLE
SONARQUBE 2
cmd
Microsoft Windows [Version 10.0.26200.9168]
(c) Microsoft Corporation. All rights reserved.
C:\Users\tanma>cd C:\Users\tanma\Downloads
C:\Users\tanma\Downloads>mkdir JenkinsProject
C:\Users\tanma\Downloads>cd JenkinsProject
C:\Users\tanma\Downloads\JenkinsProject>git init
Initialized empty Git repository in C:/Users/tanma/Downloads/JenkinsProject/.git/
C:\Users\tanma\Downloads\JenkinsProject>git add .
C:\Users\tanma\Downloads\JenkinsProject>git commit -m "Initial commit for Jenkins lab"
[master (root-commit) fc480bf] Initial commit for Jenkins lab
3 files changed, 40 insertions(+)
create mode 100644 Jenkinsfile.php
create mode 100644 index.html
create mode 100644 index.html.txt
C:\Users\tanma\Downloads\JenkinsProject>git branch -M main
C:\Users\tanma\Downloads\JenkinsProject>git remote add origin https://github.com/TanmayAhuja2006/jenkins-sample-app.git
C:\Users\tanma\Downloads\JenkinsProject>git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 787 bytes | 196.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/TanmayAhuja2006/jenkins-sample-app.git
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
C:\Users\tanma\Downloads\JenkinsProject>

index.html 1 jenkinsfile.php 1 ×

C: > Users > tanma > Downloads > jenkinsfile.php

```
pipeline {
    agent any
    stages {
        stage('Build') {
            steps {
                echo 'Building and preparing web assets...'
            }
        }
        stage('Test') {
            steps {
                echo 'Validating HTML structure...'
            }
        }
        stage('Deploy') {
            steps {
                echo 'Deploying application to local target directory...'
                bat 'if not exist "C:\\Deployments\\app" mkdir "C:\\Deployments\\app"'
                bat 'copy index.html "C:\\Deployments\\app\\index.html"'
            }
        }
    }
}
```

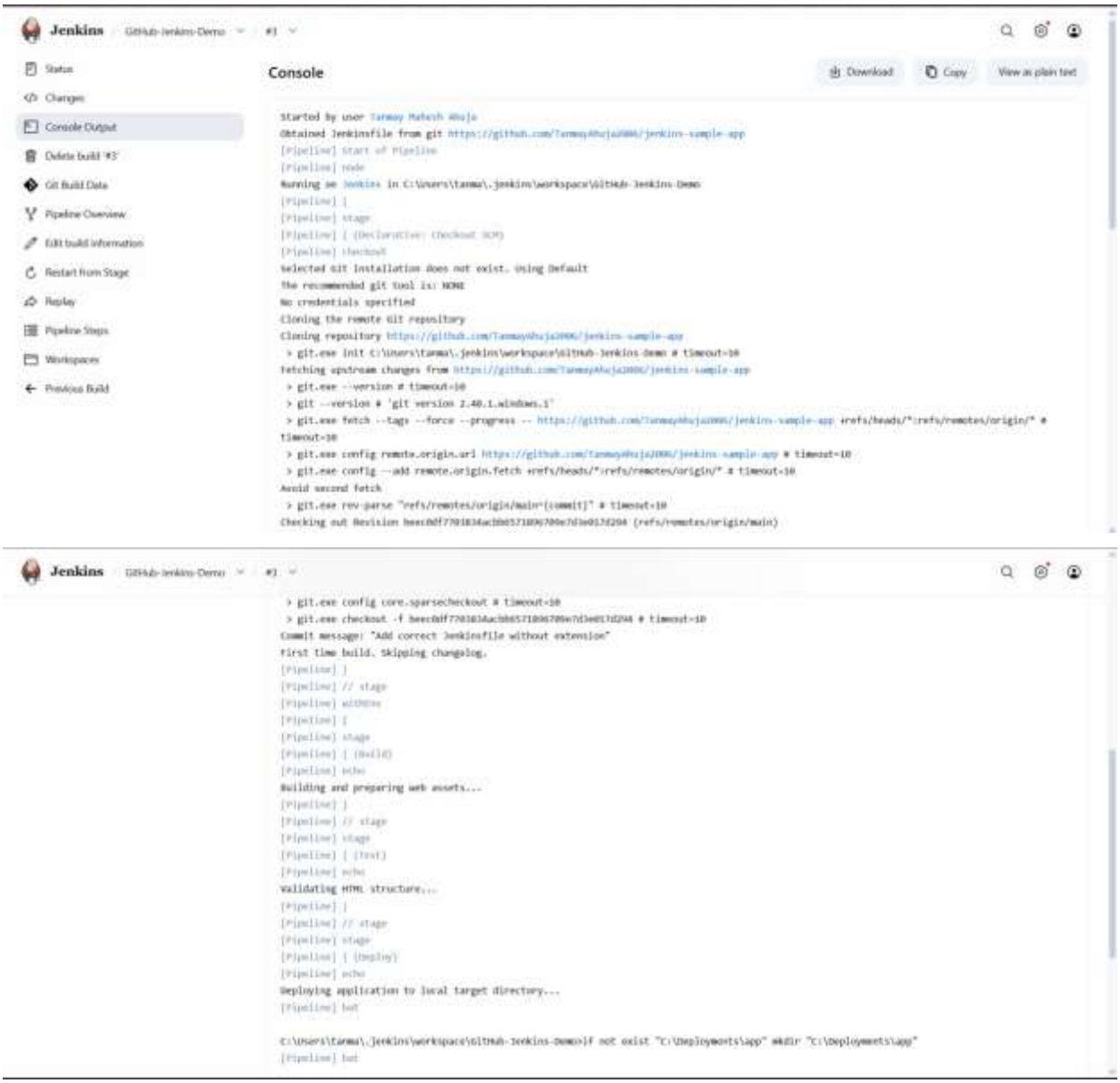
index.html 1 × jenkinsfile.php 1

C: > Users > tanma > Downloads > index.html > ...

```
<!doctype html>
<html>
  <head>
    <title>Jenkins CI/CD App</title>
  </head>
  <body>
    <h1>Welcome to Tanmay's CI/CD Deployed Web Application!</h1>
  </body>
</html>
```


Jenkins
GitHub-Jenkins-Demo
Status
Changes
Build Now
Delete Pipeline
Configure
Rename
Stages
Pipeline Syntax
GitHub Hook Log
Builds
GitHub-Jenkins-Demo
Add description
Permalinks
REST API


Jenkins
GitHub-Jenkins-Demo
#3
Status
Changes
Console Output
Delete build '#3'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
#3 (27-Aug-2026, 8:46:55 am)
Add description
Keep this build forever
Started by user
Revision:
Repository:
refs/remotes/origin/main
No changes
REST API


Jenkins
GitHub-Jenkins-Demo
#3
Status
Changes
Console Output
Delete build '#3'
Git Build Data
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Console
Download
Copy
View as plain text
Started by user Tanmay Mahesh Ahuja
Obtained Jenkinsfile from git https://github.com/TanmayAhuja2006/jenkins-sample-app
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\tanma\.jenkins\workspace\GitHub-Jenkins-Demo
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/TanmayAhuja2006/jenkins-sample-app
> git.exe init C:\Users\tanma\.jenkins\workspace\GitHub-Jenkins-Demo # timeout=10
Fetching upstream changes from https://github.com/TanmayAhuja2006/jenkins-sample-app
> git.exe --version # timeout=10
> git --version # 'git version 2.40.1.windows.1'
> git.exe fetch --tags --force --progress -- https://github.com/TanmayAhuja2006/jenkins-sample-app +refs/heads/*:refs/remotes/origin/* #
timeout=10
> git.exe config remote.origin.url https://github.com/TanmayAhuja2006/jenkins-sample-app # timeout=10
> git.exe config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision beec0df7701814ac5b6571806709e7d3e017d294 (refs/remotes/origin/main)

Jenkins
GitHub-Jenkins-Demo
#3
> git.exe config core.sparsecheckout # timeout=10
> git.exe checkout -f beec0df7701814ac5b6571806709e7d3e017d294 # timeout=10
Commit message: "Add correct Jenkinsfile without extension"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] }
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] echo
Building and preparing web assets...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] echo
Validating HTML structure...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying application to local target directory...
[Pipeline] bat
c:\Users\tanma\.jenkins\workspace\GitHub-Jenkins-Demo>if not exist "C:\Deployments\app" mkdir "C:\Deployments\app"
[Pipeline] bat



Conclusion

This experiment successfully established an end-to-end CI/CD pipeline by integrating Jenkins automation server with a GitHub source code repository. By configuring a declarative Jenkins file, the workflow seamlessly automated the software release lifecycle—ranging from source checkout and build asset preparation to HTML validation and local file deployment. The successful execution of Build #3 validates that automated pipelines eliminate manual deployment errors, ensure consistent build repeatability, and provide rapid feedback loops essential for modern software development and DevOps practices.